

Road Traffic Noise Mapping and Mitigation Study using CNOSSOS-EU

Via Ferruccio Ghinaglia in Cremona Italy

Madhav Gopi | MSc Music & Acoustic Engineering
Engineering portfolio study | 21 September 2026 | Version 1.0

Executive summary

This study uses NoiseModelling 6.0.0 and QGIS 3.44.14 to compare road-traffic noise under a baseline and two corridor interventions. A small area around Via Ferruccio Ghinaglia is evaluated at 2,667 calculation points, using OpenStreetMap geometry and explicitly assumed traffic inputs.

Reducing the assumed corridor speed from 50 to 30 km/h gives a maximum predicted Lden reduction of 2.10 dB across the outdoor receiver set. Halving corridor heavy-vehicle flows gives a maximum reduction of 1.05 dB at a different receiver. At roadside receiver R1, the corresponding reductions are 2.08 and 1.03 dB. Benefits are concentrated near the intervention corridor; whole-grid median reductions are 0.03 and 0.02 dB respectively.

These results demonstrate an environmental-noise modelling workflow and a conditional comparison of mitigation options. Traffic counts were not measured, no site validation was performed, and the results are not a certified regulatory assessment.

Study area and scenarios

The reporting area is approximately 600 by 600 m, containing roadside buildings, sheltered spaces and secondary roads. A larger OSM context extract provides surrounding roads and obstacles within the 1,000 m source search distance; it does not enlarge the reporting area. Coordinates use EPSG:32632, WGS 84 / UTM zone 32N, so distances are calculated in metres.

Scenario	Change on nine Ghinaglia road sections	Other inputs
A Baseline	50 km/h with assumed traffic flows	Reference pavement DEF
B Speed reduction	LV and HGV speeds reduced to 30 km/h	A flows retained
C HGV reduction	Heavy-vehicle flows multiplied by 0.5	A speeds and LV flows retained

LV means light vehicles; HGV means heavy goods vehicles. B and C are separate alternatives. C reduces total traffic as well as the heavy-vehicle share; displaced traffic is not reassigned to other roads. All imported geometry of the nine named secondary or secondary-link ways is included in the intervention.

Calculation sequence

OSM import → traffic assumptions → CNOSSOS-EU road emissions → propagation to a common receiver grid → Lden and scenario differences → QGIS figures. Scenario inputs, receiver identifiers and exported results are retained for reproducibility.

Inputs and calculation settings

The context dataset contains 1,267 road records, 3,680 buildings and 338 ground polygons. OSM-derived traffic flows are importer defaults, not measured Cremona counts. Numeric OSM speed tags inform 285 road records; 982 use road-class assumptions. These tags are operating-speed proxies, not verified mean speeds. All pavement codes are DEF, the CNOSSOS reference surface [1].

Parameter	Value used	Engineering interpretation
Road emission	CNOSSOS coefficient version 2	2020 coefficient set; source height 0.05 m
Baseline speed assumptions	70 / 50 / 30 / 20 km/h	Trunk / through-road / local / access roads; numeric tags take precedence
Receiver grid	10 m spacing; 4 m height	2,667 unique points; 3,321 triangles
Terrain	No DEM	Digital elevation model omitted; flat ground
Reflections	Order 1; search distance 50 m	At most one wall reflection; close-reflection exclusion 0 m
Wall absorption	0.1	10% energy absorption assumption
Diffraction	Horizontal edges on; vertical edges off	Roof-edge bending included; lateral-edge paths excluded
Source search distance	1,000 m	Finite cutoff for surrounding road contributions
Source pruning	Max Error 0 dB	Pruning disabled; does not mean zero uncertainty
Atmosphere	15 °C; 70% relative humidity	Favourable propagation probability 0.5 in all 16 directions
Ground and building heights	OSM-derived values	No field verification of ground properties or heights

The 10 m grid samples spatial variation more closely than the initial coarse diagnostic grid. A common 4 m height permits consistent scenario comparison. First-order reflections and a 1,000 m cutoff balance coverage and computation; no convergence study establishes their adequacy for a regulatory assessment. Six roads tagged 10 km/h use the model's 20 km/h low-speed approximation. Propagation settings follow the source-level workflow [2].

Assumed corridor flows in vehicles per hour

Period	LV in A B C	HGV in A B	HGV in C
Day	99	18	9
Evening	90	10	5
Night	24	1	0.5

L_{day}, L_{evening} and L_{night} are A-weighted equivalent levels for 06:00–18:00, 18:00–22:00 and 22:00–06:00. L_{den} combines their sound energies with 12/4/8-hour weights and evening/night penalties of +5/+10 dB. The penalties are weighting conventions, not physical increases in the sound.

$$L_{den} = 10 \log_{10} [(12 \cdot 10^{L_{day}/10} + 4 \cdot 10^{(L_{evening}+5)/10} + 8 \cdot 10^{(L_{night}+10)/10})/24]$$

The software's period convention is used consistently. This study does not assert compliance with Italian regulatory period definitions. A fractional hourly flow represents a time average.

Predicted baseline and speed reduction levels

Figure 1 compares baseline A with speed-reduction scenario B using identical class intervals and extent. The calculation includes all modelled roads within the search distance. A change smaller than a 5 dB colour interval may leave a polygon in the same class, so the difference maps and point results provide the clearer mitigation comparison.

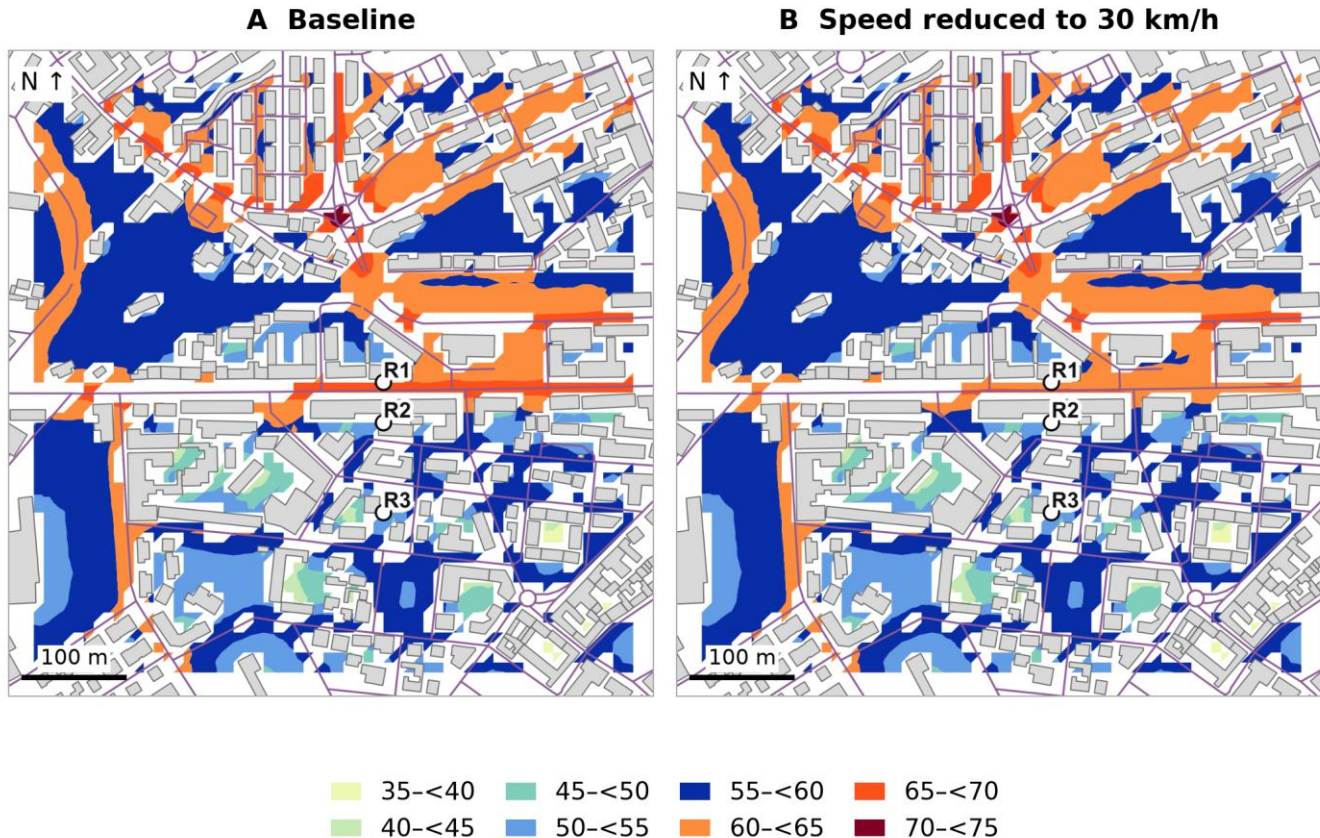


Figure 1. Predicted Lden in dB(A) at 4 m height. A: baseline. B: Ghinaglia speeds reduced from 50 to 30 km/h. Grey polygons are buildings, purple lines road centrelines, and white circles the three selected receivers. White gaps have no contour coverage. Geometry © OpenStreetMap contributors, ODbL. EPSG:32632.

Spatial interpretation

Baseline Lden ranges from 35.2 to 71.1 dB(A) across all 2,667 receivers. Elevated levels follow exposed road corridors, while some spaces behind buildings are quieter. At the same 4 m height, R1 is 65.9 dB(A), R2 is 44.4 dB(A), and R3 is 51.3 dB(A). These differences reflect the combined effects of source contributions, distances and propagation paths.

R2 is behind the first building row south of Ghinaglia. R3 is farther south but also exposed to other local roads. Its higher level than R2 shows why distance to one road alone cannot explain a network noise map. These locations do not isolate a building-screening insertion loss.

Map construction and coverage

Isobands interpolate receiver levels across the common triangles, without smoothing. ISOLVL identifies the noise class. The mesh covers 166,050 m² before building masking; outdoor difference-map coverage is 162,733 m². Blank areas have no interpolated result. Full-size A, B and C maps are retained separately [3, 4].

Mitigation effects across the study area

The difference at each matching receiver is the mitigation Lden minus baseline Lden. Negative values mean a reduction. Figure 2 interpolates these unrounded differences on the existing triangles, clips them to the study boundary and removes building footprints. It does not subtract the 5 dB classes of the absolute maps.

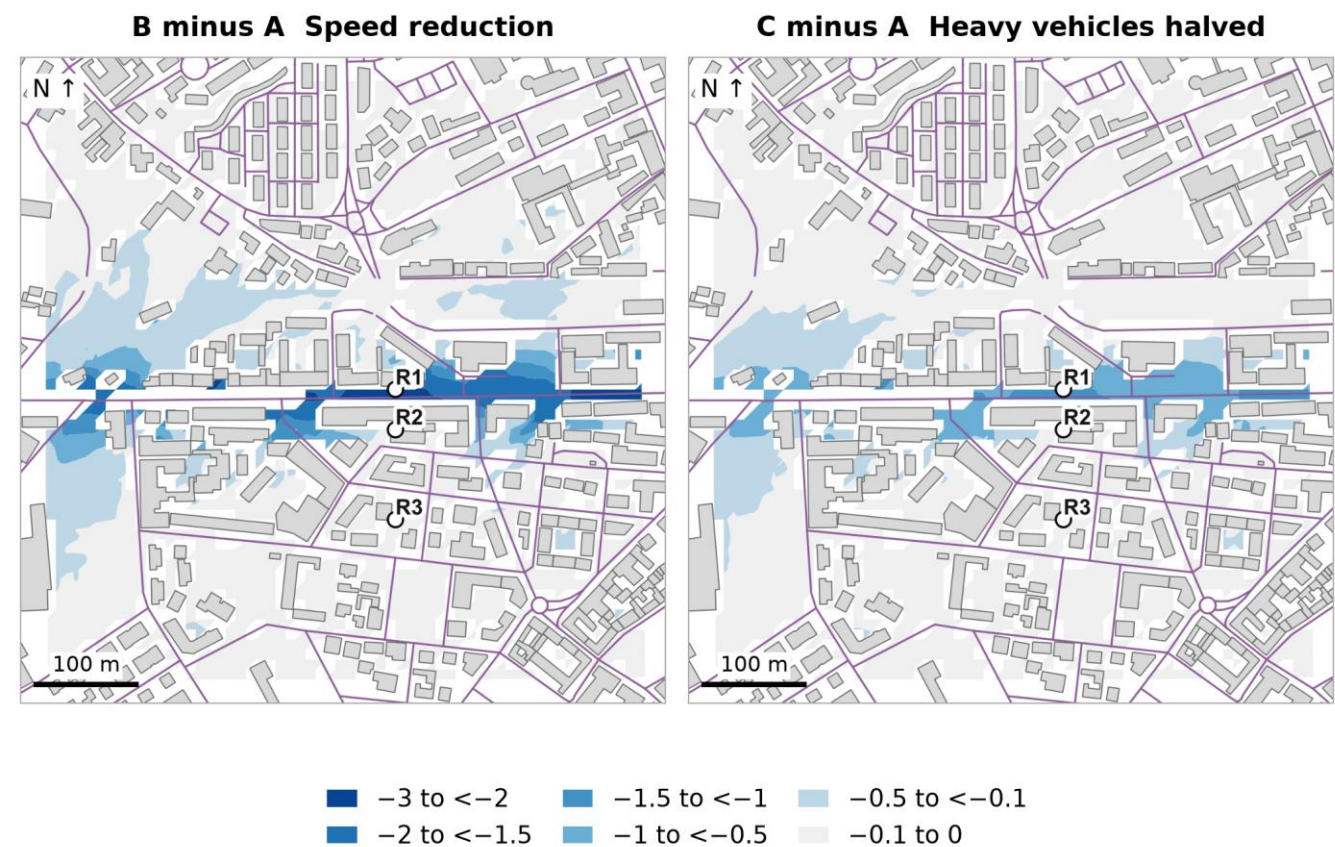


Figure 2. Change in Lden in dB, with the same scale and colour intervals for B–A and C–A. Darker blue means a larger reduction; pale grey means –0.1 to 0 dB. White gaps have no contour coverage. Grey polygons are buildings and purple lines are road centrelines. Geometry © OpenStreetMap contributors, ODbL. EPSG:32632.

Outdoor receiver statistic	B Speed reduction	C HGV reduction
Maximum reduction	2.10 dB	1.05 dB
Receiver ID at maximum	1743	1848
Median reduction	0.03 dB	0.02 dB
Receivers with reduction ≥1 dB	200 of 2,665 (7.5%)	43 of 2,665 (1.6%)

These statistics weight each of 2,665 outdoor grid points equally. Two points above low building roofs, IDs 582 and 583, are excluded from outdoor statistics. The figures are neither population-weighted nor area-weighted; 1 dB is a descriptive threshold, not an audibility or compliance criterion. The maxima occur at different receivers.

Both interventions benefit the target corridor most strongly. Their small whole-grid medians reflect the intervention’s local scope and contributions from unchanged surrounding roads. Under the chosen inputs, speed reduction produces the larger local reduction. A combined speed-and-HGV intervention was not simulated, and the two dB reductions should not simply be added.

Receiver results and traffic flow sensitivity

The three receivers were chosen to illustrate contrasting positions relative to Ghinaglia. Distances are horizontal to its modelled centreline, not facade offsets. All points use 4 m height; all contributing roads are included in their predicted levels.

Receiver	Distance m	A dB(A)	B dB(A)	C dB(A)	B-A dB	C-A dB
R1 (1805)	7.9	65.9	63.8	64.9	-2.08	-1.03
R2 (1561)	32.1	44.4	44.1	44.2	-0.35	-0.20
R3 (1012)	122.1	51.3	51.3	51.3	<0.01*	<0.01*

* Reduction smaller than 0.01 dB. Changes use unrounded results; displayed precision does not imply prediction accuracy. R1 is roadside, R2 behind the first building row, and R3 farther south.

Sensitivity to a common traffic multiplier

To test dependence on assumed traffic volume, every vehicle-category flow on every modelled road and in every period is multiplied by the same positive factor f . Speeds, fleet shares, pavement, geometry and propagation remain fixed. Source energy then scales with f [6], giving the same dB shift at every receiver:

$$L_{den}(f) = L_{den,A} + 10 \log_{10}(f)$$

Cremona | Sensitivity to assumed traffic volumes

Analytical scaling of the saved baseline: all roads, vehicle classes and periods

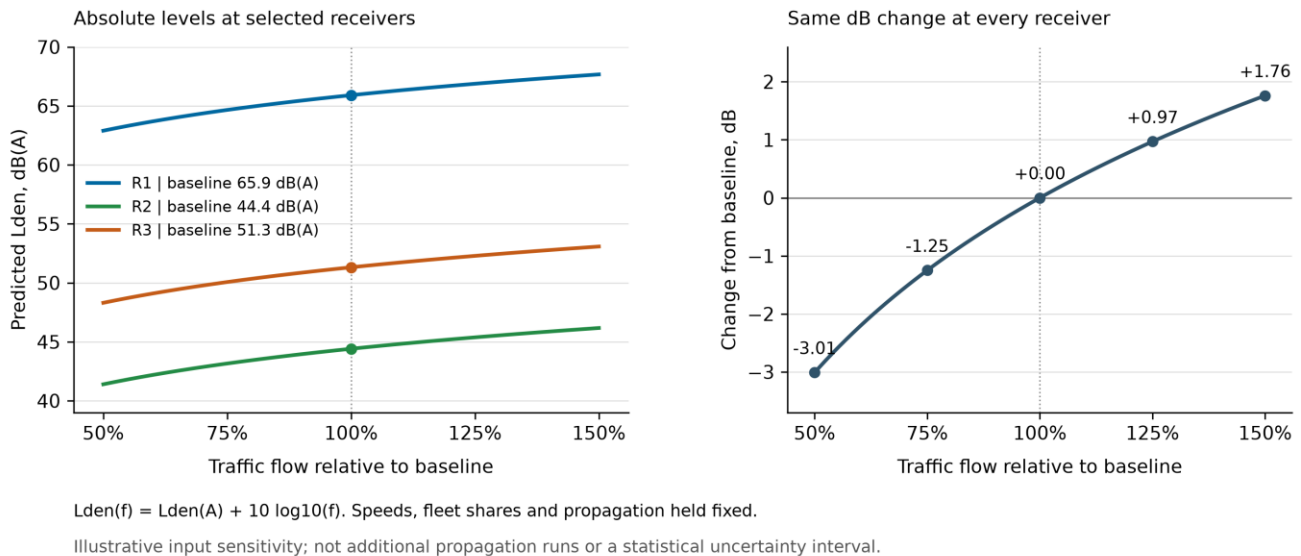


Figure 3. Analytical traffic-flow sensitivity anchored to the actual baseline receiver results. The left panel shows absolute L_{den} at R1–R3; the right shows the common change from baseline.

Half the assumed traffic gives –3.01 dB; 25% more gives +0.97 dB; 50% more gives +1.76 dB. The 50–150% range is an illustrative input test, not a statistical confidence interval. It requires no new propagation calculation because every source energy changes by the same factor.

This relationship does not substitute for the separately modelled interventions. B changes speed and therefore emission characteristics; C changes only selected heavy-vehicle flows. The sensitivity test also excludes congestion, speed changes and traffic redistribution caused by altered demand.

Limitations conclusions and reproducibility

Limits on interpretation

Traffic and validation. Flow values are assumptions from the OSM importer. OSM speed tags are imperfect proxies for operating speeds, and missing tags require class-based choices. No site traffic survey or acoustic validation measurement was performed. Absolute levels and mitigation benefits should therefore be treated as conditional model predictions.

Geometry and propagation. OSM footprints, heights, road representation and ground classes may be incomplete or inaccurate. Flat terrain omits local elevation and embankment effects. Only first-order reflections are included; meteorology and wall absorption are assumed. Grid spacing, diffraction choices, search distances and mesh gaps have not been subjected to a numerical convergence study. Railway and other non-road sources are excluded.

Scenario scope. The 30 km/h scenario assumes successful speed compliance without modelling detailed acceleration, queuing or congestion. The heavy-vehicle scenario removes corridor demand without reassigning vehicles to surrounding streets. Results describe the selected corridor and grid, not city-wide exposure or a certified Italian regulatory assessment.

Engineering conclusion

For this demonstrative study, reducing corridor speeds from 50 to 30 km/h gives a larger predicted local Lden benefit than halving heavy-vehicle flows. The roadside receiver improves by 2.08 dB under B and 1.03 dB under C. Farther and more screened locations show smaller changes because the intervention contributes less to their combined road-noise level. A practical assessment would first obtain period-specific counts, heavy-vehicle shares and operating speeds, verify geometry and terrain, and compare predictions with suitable site measurements.

Reproducibility and quality checks

All scenarios share road identifiers and geometry, building/ground context, receiver positions and propagation settings. Each export contains 10,668 records: 2,667 receivers in D, E, N and DEN. Unique receiver IDs and coordinates, CRS, heights, period coverage, valid octave-band values and Lden consistency were checked. Statistics exclude invalid -99 sentinel values. Export hashes and unrounded results are recorded in results_summary.json; comparison and sensitivity scripts reproduce the reported tables. The editable QGIS project retains five map layouts.

A complete release should retain the OSM extracts, prepared A/B/C road inputs, final receiver/triangle files, model outputs, processing scripts and QGIS project with relative paths. Source geometry is © OpenStreetMap contributors, available under the Open Database Licence; retain attribution and applicable data licence terms [5].

References

- [1] [NoiseModelling 6.0.0 documentation — road input definitions](#)
- [2] [NoiseModelling 6.0.0 documentation — Noise level from source](#)
- [3] [NoiseModelling 6.0.0 documentation — Create Isosurface](#)
- [4] [QGIS 3.44 user guide — print layouts and output](#)
- [5] [OpenStreetMap copyright and licence — openstreetmap.org/copyright](#)
- [6] [NoiseModelling source — Utils.Vperhour2NoiseLevel, CNOSSOS equation 2.2.1](#)

Online references accessed 21 September 2026. Software and result versions are fixed for this study.